AutoDD: Learning to Select Drift Detectors for Evolving Data Streams

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***Abstract*—Concept drift, the change in the distribution of streaming data over time, significantly impacts the performance of machine learning models. While numerous drift detection methods have been proposed, selecting and configuring the most suitable detector for a specific data stream remains a challenging task. Existing AutoML frameworks for streaming data that automate model selection rely on fixed drift detection strategies, leaving a critical gap in automation. This paper introduces *AutoDD* (Automated Drift Detector), a novel framework that jointly automates detector selection and hyperparameter configuration using a hierarchical learning approach. AutoDD extracts comprehensive online meta-features and employs a two-level hierarchical selection algorithm to adaptively choose both the detector and its configuration. Theoretical analysis provides sublinear regret bounds ensuring convergence to optimal performance. Comprehensive experiments on synthetic and real-world datasets demonstrate that AutoDD consistently outperforms state-of-the-art detectors while maintaining computational efficiency.**

***Index Terms*—concept drift detection, algorithm selection, contextual bandits, online meta-learning, data streams**

I. Introduction

Concept drift, where the statistical properties of streaming data evolve over time, poses a significant challenge in data stream mining [1], [2], incremental learning [3], and non-stationary learning [4]. As data distributions change, machine learning models experience degraded performance, necessitating timely detection and adaptation. This phenomenon manifests in various forms, including sudden (abrupt) drift, gradual drift, incremental drift, and recurring drift, as illustrated in Figure 1. For instance, an electricity price prediction model may encounter sudden drift when market regulations change abruptly, experience gradual drift as consumer behaviour shifts over seasons, or observe recurring drift driven by daily demand cycles [4].

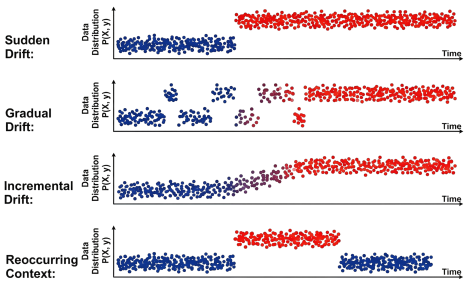


Fig. 1: Types of concept drifts.

Over the past two decades, numerous drift detection methods have been developed to address this challenge, including error rate-based approaches such as DDM [5], ADWIN [6], and ADDM [7], distribution-based methods like KSWIN [8], uncertainty-based methods such as PUDD [9], and meta-learning techniques like Meta-ADD [10]. However, a fundamental challenge remains unresolved in systematically selecting and configuring the most appropriate drift detector for a given data stream.

Automated Machine Learning (AutoML) has revolutionised the field by addressing similar selection challenges in batch learning scenarios [11]. Recent advancements have extended AutoML to streaming data, enabling automatic selection and configuration of classifiers [12], [13]. However, these systems treat drift detection as a fixed component, requiring manual specification and leaving a critical gap in the automation framework.

To bridge this gap, we introduce **AutoDD** (**A**utomated **D**rift **D**etector). This modular framework automates drift detector selection and configuration using a hierarchical learning approach, compatible with any model or AutoML pipeline.

- We propose a two-level hierarchical selection algorithm using LinUCB [14] and Thompson Sampling [15] that adaptively selects detectors and their hyperparameters based on online meta-features.
- We formalise detector selection with a theoretical analysis that provides sublinear regret bounds for the proxy selection objective optimised by the selection layer.
- Extensive evaluation demonstrates AutoDD outperforms state-of-the-art detectors while maintaining computational efficiency.

The rest of this paper is organised as follows. Section II presents the related work. Section III provides the details of our proposed AutoDD framework. Section IV describes the experimental setup. Section V presents the experimental results and analysis. Section VI discusses key design decisions and limitations. Finally, Section VII concludes the paper.

II. Related Work

Drift detection methods can be broadly categorised into three paradigms. **Error rate-based methods** monitor fluctuations in classifier error rates, triggering alarms when changes are statistically significant. DDM [5] pioneered the warning-drift framework by monitoring prediction accuracy under binomial distribution assumptions, followed by EDDM [16] which focuses on distances between errors to detect gradual drift, HDDM [17] which applies Hoeffding's inequality with exponentially weighted moving averages, and ADWIN [6] which maintains adaptive sliding windows that automatically shrink when distribution changes are detected. Page-Hinkley

[18] applies cumulative sum control charts to detect mean shifts in sequential data. Recent advances include MDDM [19], which uses McDiarmid's inequality for shorter detection delays with tighter concentration bounds, and ADDM [7], which models error rates as SETAR time series [20] to achieve lower false alarm rates. **Distribution-based methods** compare data distributions across windows by quantifying dissimilarity between new and historical data distributions [21], including approaches based on information theory [22], competence models [23], and the Kolmogorov-Smirnov test (KSWIN) [8] which detects changes in univariate feature distributions. KSWIN is a recent representative of the statistical two-sample test family, which also includes earlier detectors such as IKS [24] and WRS [25]. Fully unsupervised, model-independent variants such as the image-based detector IBDD [26] render high-dimensional windows as greyscale images and monitor their mean squared deviation, enabling label-free detection on fast, high-dimensional streams. **Uncertainty and learning-based methods** address limitations of error-rate monitoring: PUDD [9] leverages prediction confidence for earlier detection by monitoring changes in prediction uncertainty distributions, while Meta-ADD [10] employs prototypical networks to classify drift types via meta-learning on diverse drift scenarios.

Recent developments in automated algorithm selection and hyperparameter optimisation (CASH) [11] have been extended to streaming scenarios [27]. However, all existing streaming AutoML systems assume a pre-selected drift detector, requiring manual specification and leaving a critical gap in the automation framework. Despite advances in both drift detection and streaming AutoML, drift detector selection remains manual. Our work fills this gap by extending AutoML to automatic drift detector selection and configuration.

III. Automated Drift Detector

AutoDD employs an adaptive, closed-loop architecture to dynamically select and configure drift detectors. As shown in Figure 2, this architecture operates through several key steps. First, the system initialises the detector portfolio with standard hyperparameters and sets up the learning parameters. During online operation, meta-features characterising stream velocity, stationarity, and temporal patterns are continuously extracted from the incoming data stream. These meta-features serve as context for the LinUCB agent, which selects a detector from the portfolio. Thompson Sampling then chooses a hyperparameter configuration for the selected detector. The configured detector monitors the stream and triggers model adaptation upon detecting drift. Finally, performance feedback is computed based on detection outcomes, which updates both the LinUCB and Thompson Sampling components to refine future selections. This adaptive learning cycle enables AutoDD to improve its detector selections over time.

A. Algorithm Overview

AutoDD operates over sliding windows of size w. When each window fills with incoming instances, the framework executes a complete selection cycle, beginning with meta-feature extraction.

TABLE I: Detector portfolio with hyperparameter configurations.

Detector	Type	Parameter	Range	Scale	Default
ADWIN	Window-based	δ_{adwin}	[0.0001, 0.1]	log	0.002
Page-Hinkley	Window-based	θ_{ph}	[10, 100]	int	50
		δ_{ph}	[0.001, 0.01]	log	0.005
DDM	Error-based	w_{thresh}	[1.5, 3.0]	linear	2.0
		d_{thresh}	[2.0, 4.0]	linear	3.0
EDDM	Error-based	α_{eddm}	[0.90, 0.99]	linear	0.95
		β_{eddm}	[0.85, 0.95]	linear	0.90
HDDM-A	Error-based	δ_{hddm}	[0.0001, 0.01]	log	0.001
HDDM-W	Error-based	λ_{hddm}	[0.01, 0.1]	log	0.05
KSWIN	Distribution-based	α_{ks}	[0.001, 0.05]	log	0.005
		w_{ks}	[50, 500]	int	100

Meta-Feature Extraction: The meta-feature set comprises 15 features in three categories. *Statistical* features capture velocity, dimensionality, sparsity, noise level, and class imbalance; *stationarity* features capture error volatility, KL divergence, autocorrelation, mean shift, and variance shift; and *temporal* features capture trend, periodicity, change frequency, change magnitude, and recurrence score. All features are computed over a sliding window of size w using online statistics and normalised to $[0, 1]$ via running min-max normalisation to ensure scale consistency. The resulting context vector $\boldsymbol{\phi}_t \in \mathbb{R}^m$ with $m = 15$ is computed incrementally, supporting efficient per-instance updates.

Detector Portfolio: AutoDD maintains a diverse portfolio of $K = 7$ drift detectors spanning window-based, error-based, and distribution-based paradigms. Table I summarises each detector with its hyperparameters, valid ranges, sampling scales, and default values from the literature. For each detector i, we maintain C hyperparameter configurations denoted $\boldsymbol{\eta}_{i,c}$ for $c \in [C]$, the first is always the literature default, while the remaining $C - 1$ are sampled from the specified ranges using the appropriate logarithmic, linear, or integer scale.

B. Hierarchical Learning Approach

AutoDD uses a two-level hierarchical selection strategy that jointly adapts detector choice and hyperparameter configuration while maintaining computational efficiency. This decomposition avoids directly treating all $K \times C$ detector-configuration pairs as a single flat action space.

Level 1 - Detector Selection via LinUCB: We formulate detector selection as a contextual multi-armed bandit problem where context is $\boldsymbol{\phi}_t$ (meta-features), arms are K detectors, and reward is detection performance. LinUCB [14] assumes linear reward:

$$r_{i,t} = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i + \epsilon_t \tag{1}$$

where $\boldsymbol{\theta}_i \in \mathbb{R}^m$ is the unknown parameter vector for detector i, with m being the meta-feature dimensionality. The algorithm selects the detector with the highest upper confidence bound:

$$i_t = \arg\max_{i \in [K]} \left(\boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha_{\text{ucb}} \sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1} \boldsymbol{\phi}_t} \right) \tag{2}$$

where $\hat{\boldsymbol{\theta}}_i = \mathbf{A}_i^{-1} \mathbf{b}_i$ is the ridge regression estimate. Here, $\mathbf{A}_i = \lambda_{\text{reg}} \mathbf{I} + \sum_{s:i_s=i} \boldsymbol{\phi}_s \boldsymbol{\phi}_s^\top \in \mathbb{R}^{m \times m}$ is the regularised design matrix with regularisation parameter $\lambda_{\text{reg}} > 0$, and

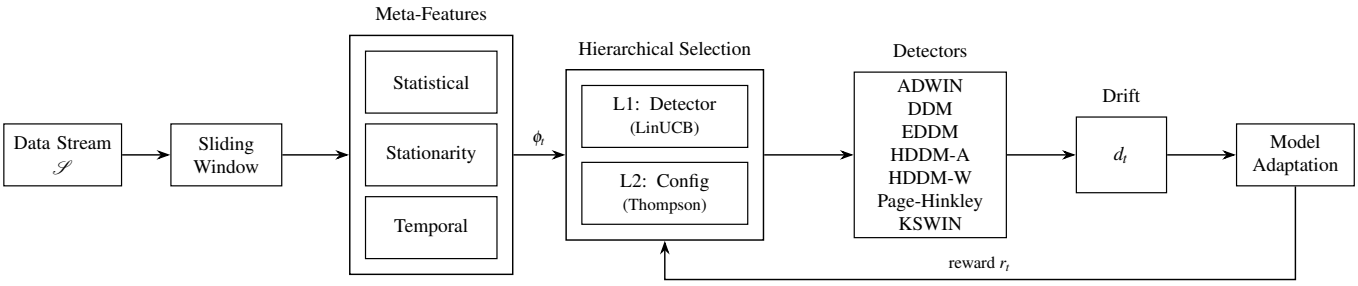


Fig. 2: The AutoDD framework: meta-feature extraction, hierarchical selection via LinUCB and Thompson Sampling, and detector portfolio with feedback loop.

$\mathbf{b}_i = \sum_{s:i_s=i} r_s \boldsymbol{\phi}_s \in \mathbb{R}^m$ is the cumulative reward-weighted context vector for detector i. The parameter $\alpha_{\text{ucb}} > 0$ controls exploration-exploitation balance.

Level 2 - Hyperparameter Selection via Thompson Sampling: Given the selected detector D_{i_t}, we adaptively configure its hyperparameters using Thompson Sampling [15]. For each configuration $c \in [C] = \{1, \ldots, C\}$ of detector i, we maintain Beta distribution parameters $(a_{i,c}, b_{i,c})$ initialised to $(1, 1)$. At each window, we sample:

$$\psi_{i_t,c} \sim \text{Beta}(a_{i_t,c}, b_{i_t,c}) \quad \forall c \in [C] \tag{3}$$

and select configuration:

$$c_t = \arg\max_c \psi_{i_t,c} \tag{4}$$

After observing reward r_t, we update:

$$a_{i_t,c_t} \leftarrow a_{i_t,c_t} + \tilde{r}_t \tag{5}$$

$$b_{i_t,c_t} \leftarrow b_{i_t,c_t} + (1 - \tilde{r}_t) \tag{6}$$

where $\tilde{r}_t = (r_t + 1)/2$ normalises the reward to $[0, 1]$.

C. Evaluation and Adaptation

Evaluating detector performance without ground-truth drift labels is challenging. Following standard practice, we use model accuracy drop as a proxy metric. The reward at time t is:

$$r_t = \begin{cases} 0.5 + \min(\Delta a_t \cdot 2, 0.5) & d_t = 1, \Delta a_t > \tau_{\text{sig}} \\ -0.3 & d_t = 1, \Delta a_t \leq \tau_{\text{sig}} \\ -0.5 - \min(\Delta a_t, 0.5) & d_t = 0, \Delta a_t > \tau_{\text{sig}} \\ 0.1 & d_t = 0, \Delta a_t \leq \tau_{\text{sig}} \end{cases} \tag{7}$$

where $d_t \in \{0, 1\}$ indicates drift detection at time t, $\Delta a_t = a_{t-w} - a_t$ is the accuracy drop between consecutive windows, and $\tau_{\text{sig}} = 0.05$ is the significance threshold. The reward function penalizes both false positives and false negatives while rewarding correct drift detections and stable periods. We note that an accuracy-based proxy couples detector quality with the downstream learner, label noise, and the evaluation window. Two design choices limit the resulting variance: the drop Δa_t is measured over a window of w instances rather than per-instance, smoothing transient noise, and the threshold τ_{sig} suppresses rewards from sub-significant fluctuations so that noise-driven dips do not reinforce spurious detections. Because the reward enters the bandit update only as a bounded scalar in $[-1, 1]$, occasional misattributed rewards perturb the posteriors gradually rather than abruptly. The regret analysis in Section III is therefore stated for this bounded proxy reward; interpreting proxy regret as true detector quality requires the proxy-fidelity assumption introduced below.

Upon drift detection, the model undergoes a complete reset. Let $\mathcal{M}_t$ denote the model state at time t. When drift is detected ($d_t = 1$), we perform:

$$\mathcal{M}_t \leftarrow \mathcal{M}_0 \tag{8}$$

where $\mathcal{M}_0$ represents the initial untrained state. The model then resumes incremental learning from subsequent instances:

$$\mathcal{M}_{t+s} \leftarrow \text{Update}(\mathcal{M}_{t+s-1}, (\mathbf{x}_{t+s}, y_{t+s})) \quad \forall s > 0 \tag{9}$$

The complete procedure is formalized in Algorithm 1.

D. Computational Complexity

AutoDD has per-window complexity of $O(wd + Km^2)$ for the incrementally maintained meta-features used in our implementation, where w is the AutoDD window size, d is the input feature dimensionality, K is the number of detectors, and $m = 15$ is the meta-feature count. The main costs are meta-feature extraction ($O(wd)$) and LinUCB matrix operations ($O(Km^2)$). Space complexity is $O(Km^2 + wd)$ for storing K covariance matrices of size $m \times m$ and the sliding window buffer.

E. Theory

We formalize the optimisation objective as a streaming detector selection problem and provide proxy-regret bounds for the hierarchical learning approach. Our analysis focuses on the selection mechanism, showing that the average proxy suboptimality vanishes under the stated contextual-bandit assumptions.

Problem Formulation. A data stream $\mathcal{S} = \{(\mathbf{x}_t, y_t)\}_{t=1}^{\infty}$ exhibits concept drift when distributions change over time:

$$\exists t, t' : P_t(\mathbf{x}, y) \neq P_{t'}(\mathbf{x}, y) \tag{10}$$

Given a detector portfolio $\mathcal{D} = \{D_1, \ldots, D_K\}$ where each D_i has configuration space $\mathcal{H}_i$, AutoDD addresses the streaming detector selection and configuration problem. Inspired by

Algorithm 1: AutoDD

Input: Stream $\mathcal{S}$, portfolio $\mathcal{D}$, window w; exploration α_{ucb}, regularisation λ_{reg}, configs per detector C
Output: Drift signals $\{d_t\}_{t=1}^{\infty}$

Initialise $\mathbf{A}_i \leftarrow \lambda_{\text{reg}}\mathbf{I}_m$, $\mathbf{b}_i \leftarrow \mathbf{0}$ for all $i \in [K]$;
Initialise $a_{i,c} \leftarrow 1$, $b_{i,c} \leftarrow 1$ for all $i \in [K], c \in [C]$;
Initialise model $\mathcal{M}_0$, sliding window $W \leftarrow \emptyset$;
foreach *instance* $(\mathbf{x}_t, y_t) \in \mathcal{S}$ **do**
  $W \leftarrow W \cup \{(\mathbf{x}_t, y_t)\}$;
  if $|W| = w$ **then**
    $\boldsymbol{\phi}_t \leftarrow \textsc{Extract}(W)$; `// meta-features`
    for $i = 1$ **to** K **do**
      $\text{UCB}_i \leftarrow \boldsymbol{\phi}_t^\top \hat{\boldsymbol{\theta}}_i + \alpha_{\text{ucb}} \sqrt{\boldsymbol{\phi}_t^\top \mathbf{A}_i^{-1} \boldsymbol{\phi}_t}$;
      `//` $\hat{\boldsymbol{\theta}}_i = \mathbf{A}_i^{-1}\mathbf{b}_i$
    $i_t \leftarrow \arg\max_i \text{UCB}_i$; `// select detector`
    for $c = 1$ **to** C **do**
      Sample $\psi_c \sim \text{Beta}(a_{i_t,c}, b_{i_t,c})$;
    $c_t \leftarrow \arg\max_c \psi_c$; `// select configuration`
    $d_t \leftarrow D_{i_t}(\boldsymbol{\eta}_{i_t,c_t}).\textsc{Detect}(W)$;
    if $d_t = 1$ **then**
      $\mathcal{M}_t \leftarrow \mathcal{M}_0$; `// reset model`
    Compute reward r_t using Eq. (7);
    $\mathbf{A}_{i_t} \leftarrow \mathbf{A}_{i_t} + \boldsymbol{\phi}_t\boldsymbol{\phi}_t^\top$; $\mathbf{b}_{i_t} \leftarrow \mathbf{b}_{i_t} + r_t\boldsymbol{\phi}_t$;
    $\tilde{r}_t \leftarrow (r_t + 1)/2$;
    $a_{i_t,c_t} \leftarrow a_{i_t,c_t} + \tilde{r}_t$; $b_{i_t,c_t} \leftarrow b_{i_t,c_t} + (1 - \tilde{r}_t)$;
    $W \leftarrow \emptyset$;

the CASH formulation [11], we jointly optimise the detector choice and its hyperparameters:

$$(D_t^*, \boldsymbol{\eta}_t^*) = \arg\max_{D_i \in \mathcal{D}, \boldsymbol{\eta} \in \mathcal{H}_i} \mathbb{E}[\mathcal{M}(D_i(\boldsymbol{\eta}), \mathcal{S}_t) \mid \boldsymbol{\phi}_t] \tag{11}$$

where $\boldsymbol{\phi}_t$ is the stream context, $\mathcal{S}_t$ denotes the stream segment at time t, $\mathbb{E}[\cdot]$ is the expectation operator, and $\mathcal{M}$ denotes the downstream model performance.

Assumptions. We adopt three standard assumptions from the contextual bandit literature [28], [29], together with a fidelity condition (A4) that links our accuracy-based proxy reward to true detection quality:

- **(A1) Bounded features:** $\|\boldsymbol{\phi}_t\|_2 \le L$ for all t, satisfied by $[0,1]$ normalisation with $L = \sqrt{m}$.
- **(A2) Linear reward:** For each detector i, there exists $\boldsymbol{\theta}_i^* \in \mathbb{R}^m$ with $\|\boldsymbol{\theta}_i^*\|_2 \le S$ such that $\mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t] = \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^*$, where $S > 0$ is a constant upper bound on the parameter norms.
- **(A3) Sub-Gaussian noise:** Reward noise is σ_{noise}-sub-Gaussian, satisfied with $\sigma_{\text{noise}} = 1$ for bounded rewards in $[-1, 1]$.
- **(A4) Proxy fidelity:** The windowed accuracy-drop reward (Eq. (7)) is, in expectation, monotone in true detection quality, so that a high proxy-reward detector-configuration pair is also high quality under the objective in Eq. (11). The bounds below are stated with respect to this proxy objective; reading proxy regret as evidence of convergence towards a high-quality detector additionally relies on A4 as a modelling assumption.

These assumptions characterise the *selection layer* under the standard linear contextual-bandit abstraction; they are not claims about the full loop spanning detection, adaptation, and learning. A2 in particular treats the windowed reward as linear in the meta-features in expectation, an approximation we do not prove directly but examine empirically through the selection-accuracy trend in Table VII.

Remark (stationarity under drift). Although the data distribution P_t is non-stationary by construction, A2 requires only that the *conditional* reward map $\boldsymbol{\phi}_t \mapsto \mathbb{E}[r_{i,t} \mid \boldsymbol{\phi}_t]$ be stationary, not the marginal P_t. This is the intended role of the meta-features: $\boldsymbol{\phi}_t$ is designed to encode the current stream regime (velocity, stationarity, and temporal structure), so changes in P_t can be represented through the context rather than through time-varying $\boldsymbol{\theta}_i^*$. A detector well matched to a recurring regime should earn comparable reward whenever that regime recurs, keeping $\boldsymbol{\theta}_i^*$ approximately fixed across time. When this abstraction is imperfect, i.e., the reward map itself drifts, sliding-window or discounted LinUCB variants could instead be analysed using dynamic-regret tools with a variation budget $B_T = \sum_t \|\boldsymbol{\theta}_{i,t}^* - \boldsymbol{\theta}_{i,t-1}^*\|_2$ [29]. The monotonic rise of proxy-oracle selection accuracy in Table VII is consistent with the stationary approximation on the streams we study.

The following result gives the detector-selection proxy regret for LinUCB under the stationary linear abstraction.

Theorem 1 (Level 1 Regret). Under assumptions A1–A3, with time horizon T, confidence parameter $\delta_{\text{conf}} \in (0, 1)$, and exploration parameter

$$\alpha_{\text{ucb}} = \sigma_{\text{noise}} \sqrt{m \log(KT/\delta_{\text{conf}})} + S, \tag{12}$$

LinUCB detector selection achieves, with probability at least $1 - \delta_{\text{conf}}$,

$$R_1(T) = \sum_{t=1}^{T} \left(\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \right) = O(\sqrt{KmT \log T}) \tag{13}$$

Proof Sketch. We analyse the disjoint-arm LinUCB algorithm [14] using self-normalised confidence bounds [28]. The self-normalised martingale inequality places each $\boldsymbol{\theta}_i^*$, with probability $1 - \delta_{\text{conf}}$, in the confidence ellipsoid

$$\|\boldsymbol{\theta}_i^* - \hat{\boldsymbol{\theta}}_i\|_{\mathbf{A}_i} \le \alpha_{\text{ucb}}. \tag{14}$$

On this event the UCB rule bounds the instantaneous regret as

$$\max_i \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_i^* - \boldsymbol{\phi}_t^\top \boldsymbol{\theta}_{i_t}^* \;\le\; 2\alpha_{\text{ucb}} \|\boldsymbol{\phi}_t\|_{\mathbf{A}_{i_t}^{-1}}. \tag{15}$$

Summing over t, applying the elliptical potential lemma [30] to each of the K per-detector estimators, and combining the potentials by Cauchy–Schwarz contributes the $\sqrt{K}$ factor and gives $R_1(T) = O(\sqrt{KmT \log T})$, treating the confidence

radius $\alpha_{\text{ucb}} = \tilde{O}(\sqrt{m})$ as constant; retaining it gives the dimension-explicit $O(m\sqrt{KT})$. □

Thompson Sampling subsequently balances exploration and exploitation for hyperparameter configuration.

Proposition 2 (Level 2 Proxy Regret). For Thompson Sampling on C configurations per detector under the standard Bernoulli-sampling reduction for bounded rewards, with $N_i(T)$ denoting the number of times detector $i \in [K]$ is selected over T rounds, the expected cumulative proxy regret satisfies:

$$R_2(T) = \sum_{i=1}^{K} O(\sqrt{C \cdot N_i(T) \log N_i(T)}) \tag{16}$$

Proof Sketch. Conditioned on detector i being played, hyperparameter selection is a C-armed bandit with rewards normalised to $\tilde{r}_t = (r_t + 1)/2 \in [0, 1]$. Under the Bernoulli-sampling reduction with Beta posteriors, Thompson Sampling attains expected regret $O(\sqrt{CN_i(T) \log N_i(T)})$ over the $N_i(T)$ rounds detector i is played [31]; summing over the K detectors with $\sum_i N_i(T) = T$ gives the bound. The fractional Beta update (Eq. (5)) is the conditional expectation of this reduction, used as a stable implementation heuristic. □

Under the same proxy-reward abstraction, the combined regret remains sublinear, implying that average per-round proxy suboptimality vanishes over time.

Corollary (Combined Proxy Regret). Let (i_t^*, c_t^*) denote the per-context proxy-oracle detector-configuration pair at round t. If the hierarchical proxy objective decomposes into detector-selection and within-detector configuration terms, then the total regret can be written as $R(T) = R_1(T) + R_2(T)$, where $R_1(T)$ captures detector-selection suboptimality and $R_2(T)$ captures configuration-selection suboptimality. The combined bound is:

$$R(T) = O(\sqrt{KmT \log T} + \sqrt{CKT \log T}) \tag{17}$$

Proof Sketch. Telescoping the per-round gap through the best configuration of the *selected* detector splits it into detector- and configuration-selection terms, bounded by Theorem 1 and Proposition 2 respectively. For the latter, Cauchy–Schwarz gives

$$\sum_{i=1}^{K} \sqrt{N_i(T)} \;\le\; \sqrt{K \textstyle\sum_{i=1}^{K} N_i(T)} \;=\; \sqrt{KT}, \tag{18}$$

so $R_2(T) = O(\sqrt{CKT \log T})$; adding $R_1(T)$ yields the result. □

Thus, under the proxy-reward and decomposition assumptions, the average proxy regret vanishes:

$$\lim_{T \to \infty} \frac{R(T)}{T} = 0 \tag{19}$$

Empirical support for this learning behaviour is provided in Table VII, where AutoDD's proxy-oracle selection accuracy improves from 69.4% in the first quarter of the stream to 93.1% by stream end. This trend is consistent with the linear reward approximation on the evaluated streams, but does not by itself prove the approximation. As a concrete reference point, inverting the dominant $O(\sqrt{KmT \log T})$ term for our portfolio ($K = 7$, $m = 15$) to reach an average proxy suboptimality on the order of 10^{-2} requires a horizon of roughly $T \approx 5 \times 10^5$ windows, well beyond the length of any individual stream we evaluate. The convergence in Table VII therefore reflects the early, steep portion of this regret curve rather than its asymptotic tail.

TABLE II: Dataset characteristics (Multiple* = Sudden, Gradual, Incremental, and Recurring drift types)

Dataset	Type	Instances	Features	Drift Type
SEA	Synthetic	500,000	3	Multiple*
HYP	Synthetic	500,000	10	Multiple*
AGR	Synthetic	100,000	9	Multiple*
RBF	Synthetic	200,000	10	Multiple*
RTG	Synthetic	100,000	30	Multiple*
Elec	Real-world	45,312	8	Unknown
Weather	Real-world	18,159	8	Unknown
Spam	Real-world	4,601	57	Unknown
Airlines	Real-world	539,383	7	Unknown
Poker	Real-world	829,201	10	Unknown

IV. Experiments

We evaluate AutoDD on ten benchmark datasets comprising five synthetic and five real-world streams, with characteristics detailed in Table II. To isolate drift detection performance from model selection effects, all methods use two fixed base models: GaussianNB and Hoeffding Adaptive Tree (HAT) [32]. These two learners represent fundamentally different model families (probabilistic and tree-based) and are the standard benchmark base learners used in the data stream mining literature [2], [25]. We compare AutoDD against twelve baselines: a no-detector baseline, seven portfolio detectors including ADWIN, DDM, EDDM, HDDM-A, HDDM-W, Page-Hinkley, KSWIN, and four state-of-the-art methods including MDDM [19], ADDM [7], PUDD [9], and Meta-ADD [10]. All detectors use recommended hyperparameter settings from the literature. The portfolio includes one recent representative per detector family rather than every variant (for example, KSWIN for the statistical two-sample test family); because AutoDD's selection mechanism is detector-agnostic, a larger portfolio extends only the action set, not the method.

We measure prequential accuracy as the percentage of correct predictions using test-then-train evaluation [2]. For synthetic datasets with known drift points, we report F1-score computed by matching detected drifts to ground truth using a tolerance window of $\tau_{\text{tol}} = 100$ instances following standard practice [6]. For real-world datasets, we report the number of detected drifts (DN). These metrics are complementary by design, as prequential accuracy captures the downstream task impact of a detector's adaptation decisions, whereas the drift-detection F1-score scores precision and recall against ground-truth drift points and thus credits timely, correct detections while penalizing both missed drifts and spurious alarms that accuracy alone would not separate. On real-world streams the drift points are unknown, so F1 cannot be computed

and we instead report DN alongside accuracy; following the benchmarking guidance of Souza et al. [33], we interpret DN relative to the consensus detection range across methods rather than as an absolute target, treating values far above or below this range as over- and under-detection respectively. AutoDD uses all 15 meta-features with window size $w = 100$, exploration parameter $\alpha_{\text{ucb}} = 1.0$, regularisation parameter $\lambda_{\text{reg}} = 1.0$, and $C = 5$ hyperparameter configurations per detector. Results are averaged over 10 runs with random seeds uniformly sampled from the range [1, 100]. Implementation uses Python with River [34] on a MacBook Pro with Apple M2 Pro and 16GB RAM.[1]

V. Results

We conduct a comprehensive evaluation of AutoDD against classical detectors and state-of-the-art methods across both base models.

A. Synthetic Datasets

Table III presents results on synthetic benchmarks with known drift points. AutoDD achieves the highest mean accuracy across both base models, indicating robust performance across the two learning paradigms evaluated. With GaussianNB, AutoDD attains 81.9% mean accuracy, while with HAT it reaches 84.0%, exceeding the compared state-of-the-art detectors in both settings. Meta-ADD ranks as the second-best method, achieving 80.9% with GaussianNB and 81.4% with HAT. AutoDD shows particular strength on datasets with abrupt drift patterns, where its ability to switch between detectors yields substantial benefits. On the SEA dataset, AutoDD achieves 93.1% accuracy with GaussianNB and 96.7% with HAT, outperforming all baselines in this setting. Similar improvements are observed on the AGR dataset, with 78.5% and 82.4% accuracy, respectively. Meta-ADD performs competitively on gradual drift scenarios due to its learned drift-type representations. Notably, fixed detectors exhibit inconsistent performance across different drift types, with HDDM-W achieving strong results on some datasets but underperforming on others, motivating adaptive detector selection. A detailed per-drift-type breakdown (sudden, incremental, gradual, recurring) is provided in Appendix A (Table VIII).

B. Real-world Datasets

Table IV presents results on real-world benchmarks where drift patterns are unknown and potentially more complex than synthetic scenarios. AutoDD achieves the highest mean accuracy with both base models. With GaussianNB, AutoDD attains 79.1% mean accuracy, improving to 83.1% with HAT, exceeding the compared state-of-the-art detectors on the mean metric. ADDM emerges as the second-best performer with 75.1% accuracy using GaussianNB and 77.8% with HAT (tied with PUDD under HAT). AutoDD's improvements appear across all five real-world datasets, suggesting that its adaptive selection mechanism transfers across the evaluated domains. On the Electricity dataset, AutoDD achieves 89.6% with GaussianNB and 93.8% with HAT. On Weather, it reaches 80.0% and 85.1% respectively, while on Spam it attains 97.9% and 99.0%. AutoDD also maintains a detection count within the consensus range of the compared methods, avoiding the most extreme over- and under-detection patterns. On the Airlines dataset, AutoDD detects 118 drifts, compared to HDDM-W's 269 detections and ADWIN's 76 detections, indicating more moderate sensitivity in this benchmark. Figure 3 traces prequential accuracy over time on SEA and Poker for both base learners. AutoDD (bold curve) tracks or leads the baselines throughout and recovers quickly after accuracy dips, consistent with its lower detection delay (Table VI).

[1]Code: https://anonymous.4open.science/r/AutoDD-BCCF

C. Statistical Significance

To validate the observed performance differences, we conduct statistical significance analysis using the Friedman test with Nemenyi post-hoc analysis at $\alpha = 0.05$. Figures 4a and 4b present the Critical Difference diagrams for GaussianNB and HAT, respectively. AutoDD achieves the best average rank under both base models, with Meta-ADD, ADDM, PUDD, and MDDM following in subsequent ranks. The No Detector baseline performs worst, supporting the value of drift detection in these streaming benchmarks.

D. Ablation Study

To understand the contribution of each component, we systematically remove AutoDD components and evaluate the impact on performance using GaussianNB as the base model. Table V presents results averaged across all ten datasets. The hierarchical selection mechanism is important for AutoDD's performance. Removing Level 1 LinUCB detector selection and replacing it with random selection causes the largest performance drop, reducing accuracy from 80.5% to 62.5%, an 18.0% decrease. This degradation indicates that detector-stream matching is a major contributor to effective drift detection in our evaluation. Removing Level 2 Thompson Sampling hyperparameter configuration reduces accuracy to 75.3%, showing that adaptive tuning contributes beyond fixed default configurations. All three meta-feature categories contribute to the overall performance. Statistical features have the largest impact, with their removal reducing accuracy by 9.3%. Stationarity features contribute 7.0% to performance, while temporal features account for 6.3%. Figure 5 visualizes AutoDD's adaptive behaviour on the Hyperplane stream (a single run, $w = 100$), showing how the LinUCB layer continually re-selects detectors as the stream's meta-features evolve rather than committing to a single detector, while the windowed prequential error stays bounded and recovers after detected drifts.

E. Sensitivity Study

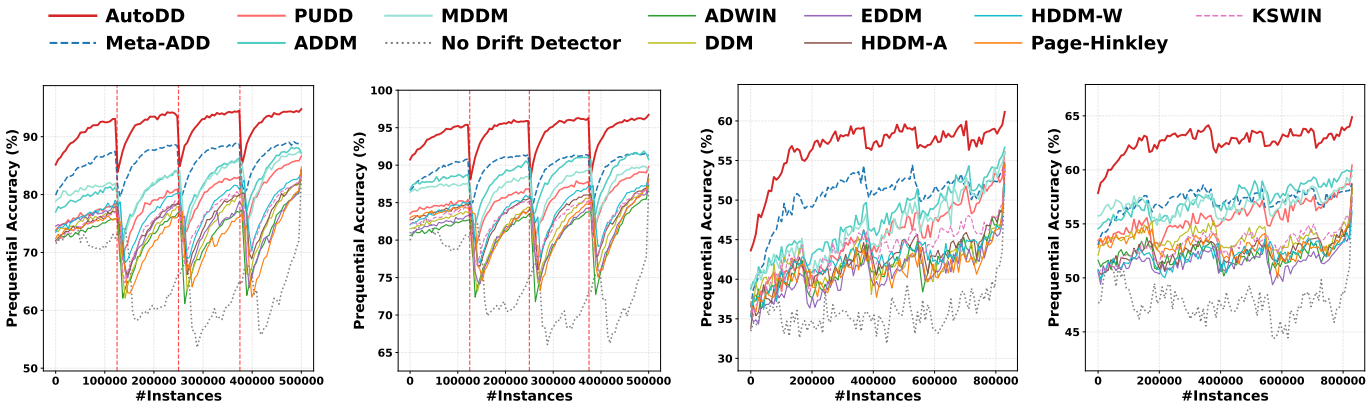
We analyse AutoDD's sensitivity to key hyperparameters using GaussianNB as the base model (Figure 6). Among the tested values, window size $w = 100$ achieves the best selection

TABLE III: Results on synthetic datasets with both base models. **Bold**: best, Underline: second-best.

	GaussianNB (GNB)												Hoeffding Adaptive Tree (HAT)											
	SEA		HYP		AGR		RBF		RTG		Mean		SEA		HYP		AGR		RBF		RTG		Mean	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
No Detector	83.8	0.40	76.4	0.34	78.8	0.41	50.4	0.09	58.2	0.10	69.5	0.27	86.2	0.42	79.1	0.36	81.3	0.43	53.1	0.11	60.8	0.12	72.1	0.29
ADWIN	84.0	0.61	81.9	0.00	71.5	0.51	69.6	0.00	64.5	0.00	74.3	0.22	88.1	0.64	85.5	0.06	74.6	0.55	73.3	0.07	68.3	0.11	77.9	0.29
DDM	83.8	0.00	86.1	0.63	71.6	0.49	71.3	0.64	64.4	0.00	75.4	0.35	88.0	0.10	89.0	0.67	74.7	0.52	74.6	0.69	68.1	0.13	78.9	0.42
EDDM	84.3	0.40	85.7	0.63	65.3	0.24	67.7	0.38	65.0	0.40	73.6	0.41	88.0	0.46	88.6	0.68	69.2	0.29	71.8	0.42	68.8	0.47	77.3	0.46
HDDM-A	84.6	0.69	81.5	0.00	72.2	0.63	69.7	0.00	64.4	0.00	74.5	0.26	88.3	0.64	86.8	0.06	75.1	0.63	73.8	0.07	69.1	0.11	78.6	0.30
HDDM-W	84.4	0.66	86.6	0.71	73.1	0.67	70.1	0.51	65.8	0.56	76.0	0.62	88.4	0.65	89.5	0.72	75.9	0.67	73.9	0.53	69.7	0.58	79.5	0.63
Page-Hinkley	84.7	0.70	81.5	0.00	71.5	0.49	69.4	0.00	64.5	0.00	74.3	0.24	88.2	0.65	86.0	0.06	74.8	0.52	73.5	0.07	68.5	0.11	78.2	0.28
KSWIN	83.7	0.35	85.7	0.65	71.5	0.50	70.0	0.61	65.2	0.39	75.2	0.50	88.0	0.42	88.3	0.66	74.5	0.53	73.7	0.63	68.9	0.43	78.7	0.53
MDDM	86.7	0.65	87.2	0.68	72.3	0.71	69.8	0.56	66.1	0.56	76.4	0.63	89.1	0.67	90.3	0.75	75.5	0.75	73.8	0.60	69.8	0.61	78.4	0.68
ADDM	87.7	0.83	88.7	0.73	73.6	0.67	70.9	0.62	67.6	0.71	77.7	0.71	89.8	0.86	91.9	0.77	76.6	0.72	75.0	0.69	71.3	0.77	79.3	0.76
PUDD	86.2	0.81	89.5	0.84	73.7	0.72	73.4	0.72	67.8	0.71	78.1	0.76	89.8	0.83	92.6	0.83	77.1	0.76	76.4	0.74	71.9	0.74	79.7	0.78
Meta-ADD	88.9	0.87	90.8	**0.91**	74.1	0.79	**78.0**	0.70	**72.9**	0.76	80.9	0.81	90.9	0.91	92.5	0.92	77.1	0.81	**84.8**	0.74	**80.4**	0.79	81.4	0.83
AutoDD	**93.1**	**0.90**	**94.6**	**0.91**	**78.5**	**0.84**	74.3	**0.81**	69.2	**0.84**	**81.9**	**0.86**	**96.7**	**0.93**	**98.4**	**0.93**	**82.4**	**0.90**	75.8	**0.84**	70.9	**0.88**	**84.0**	**0.87**

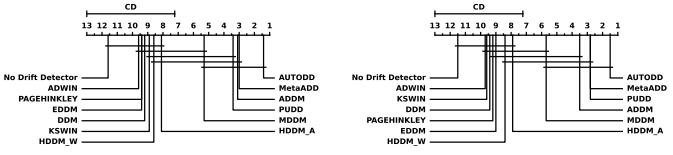
TABLE IV: Results on real-world datasets with both base models. DN = detected drifts. **Bold**: best, Underline: second-best.

	GaussianNB (GNB)												Hoeffding Adaptive Tree (HAT)											
	Elec		Weather		Spam		Airlines		Poker		Mean		Elec		Weather		Spam		Airlines		Poker		Mean	
	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN	Acc	DN
No Detector	73.2	0	68.9	0	77.2	0	59.3	0	54.6	0	66.6	0	76.5	0	71.4	0	80.1	0	62.1	0	56.9	0	69.4	0
ADWIN	79.2	22	69.3	7	92.1	2	62.5	76	54.3	3	71.5	22.0	82.4	22	73.1	9	94.6	2	66.3	76	57.7	3	74.8	22.4
DDM	82.3	22	70.6	5	91.8	2	61.8	20	54.2	3	72.1	10.4	85.5	22	73.5	7	93.9	2	65.2	20	58.0	3	75.2	10.8
EDDM	82.4	22	73.8	9	92.9	2	61.3	182	53.6	4	72.8	43.8	85.2	22	77.1	9	94.8	2	64.6	182	57.8	4	75.9	43.8
HDDM-A	82.7	22	72.2	9	92.5	2	62.6	114	54.6	6	72.9	30.6	86.3	22	75.8	9	94.7	2	66.1	114	58.5	6	76.3	30.6
HDDM-W	81.9	22	72.6	9	91.9	1	61.0	269	52.3	394	71.9	139.0	85.1	22	76.0	9	94.5	1	64.9	269	56.8	404	75.5	141.0
Page-Hinkley	78.0	16	69.4	6	91.8	1	62.8	64	54.3	17	71.3	20.8	81.7	16	72.8	6	94.2	1	66.5	64	57.5	17	74.5	20.8
KSWIN	81.1	22	68.0	9	92.6	2	61.6	269	53.7	266	71.4	113.6	84.3	22	71.2	9	94.9	2	65.4	269	57.0	266	74.6	113.6
MDDM	83.9	22	74.8	9	92.8	2	62.2	110	56.1	190	74.0	66.6	86.5	22	78.0	9	94.6	2	66.0	110	60.0	190	77.0	66.6
ADDM	84.8	22	76.3	9	93.9	2	63.9	74	56.5	177	75.1	56.8	87.3	22	79.2	9	95.8	2	66.9	74	59.6	177	77.8	56.8
PUDD	84.4	22	75.4	9	93.2	2	64.0	117	56.2	81	74.6	46.2	87.5	22	78.6	9	95.6	2	67.7	117	59.7	81	77.8	46.2
Meta-ADD	85.4	22	74.9	9	94.5	2	63.8	139	55.5	182	74.8	70.8	87.9	22	78.2	9	95.9	2	66.5	139	59.6	182	77.6	70.8
AutoDD	**89.6**	22	**80.0**	9	**97.9**	2	**68.2**	118	**59.7**	188	**79.1**	67.8	**93.8**	22	**85.1**	9	**99.0**	2	**72.6**	118	**65.1**	188	**83.1**	67.8



(a) SEA (GNB) (b) SEA (HAT) (c) Poker (GNB) (d) Poker (HAT)

Fig. 3: Prequential accuracy over time. SEA (synthetic) and Poker (real-world) datasets with GaussianNB and HAT base models.



(a) GaussianNB (b) HAT

Fig. 4: Critical Difference diagrams (Friedman-Nemenyi, $\alpha = 0.05$). AutoDD achieves the best average rank under both base models.

TABLE V: Ablation study results (accuracy averaged across all ten datasets; F1 over the five synthetic streams with known drift points).

Configuration	Accuracy	F1	ΔAcc
Full AutoDD	**80.5%**	**0.64**	–
w/o Temporal	74.2%	0.59	-6.3%
w/o Statistical	71.2%	0.57	-9.3%
w/o Stationarity	73.5%	0.59	-7.0%
w/o LinUCB (Level 1)	62.5%	0.50	-18.0%
w/o Thompson (Level 2)	75.3%	0.60	-5.2%

accuracy, balancing statistical stability with drift responsiveness. Smaller windows suffer from high variance, while larger windows cause delayed drift response. For the exploration parameter, $\alpha_{\text{ucb}} = 1.0$ achieves the best performance among the tested values, with lower values causing exploitation bias and higher values leading to excessive exploration. The regularisation parameter λ_{reg} shows robust performance across the range from 0.5 to 2.0, with $\lambda_{\text{reg}} = 1.0$ performing best in this sweep. This stability indicates that AutoDD is not overly sensitive to hyperparameter choices around the recommended

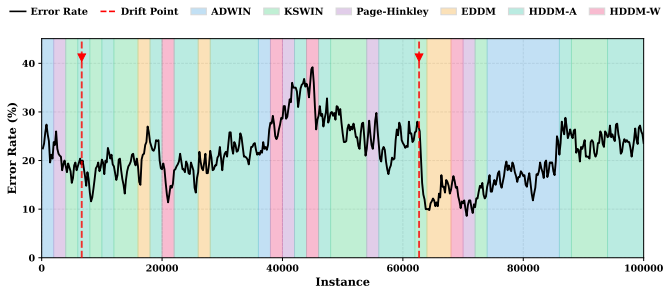
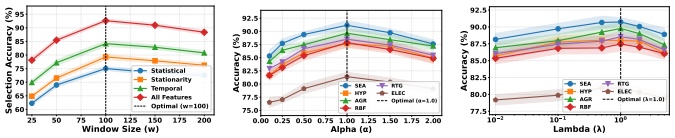
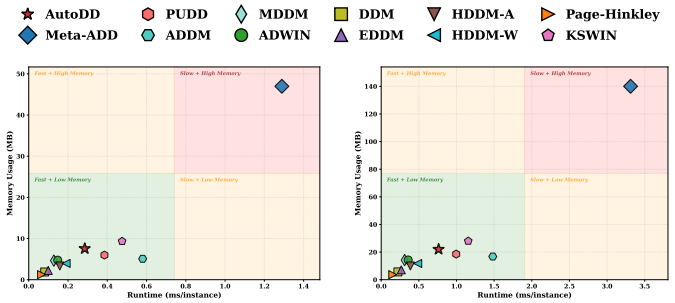


Fig. 5: AutoDD detector selection on the Hyperplane stream (single run, seed 42, w=100). Coloured bands show the dominant LinUCB-selected detector per 2,000-instance block, the black curve is the windowed prequential error rate, and dashed red lines mark drifts flagged by the active detector.



(a) w (b) α_{ucb} (c) λ_{reg}

Fig. 6: Hyperparameter sensitivity analysis showing selection accuracy across different values of (a) window size w, (b) exploration parameter α_{ucb}, and (c) regularisation parameter λ_{reg}.



(a) GaussianNB (b) HAT

Fig. 7: Runtime vs. memory trade-off. AutoDD lies in the Fast + Low Memory quadrant under both base models.

defaults.

F. Computational Efficiency

Figures 7a and 7b compare runtime and memory usage across all methods for GaussianNB and HAT, respectively. AutoDD falls within the desirable *Fast + Low Memory* quadrant under both base models, demonstrating that its hierarchical selection mechanism introduces only modest overhead. Compared to state-of-the-art methods such as Meta-ADD and PUDD, AutoDD achieves competitive efficiency while delivering better detection performance, making it practical for high-velocity streaming scenarios.

G. Detection Delay

Beyond accuracy, timely drift detection is critical for rapid model adaptation. Table VI reports mean detection delay (in instances) across synthetic datasets with known drift points. AutoDD achieves the lowest mean delay of 38.4 instances, outperforming PUDD (41.8), Meta-ADD (44.7), ADDM (46.8), and all classical detectors in this comparison. This improvement is consistent with AutoDD selecting more responsive detectors in drift-prone regimes while avoiding over-sensitive detectors during stable periods.

TABLE VI: Mean detection delay (instances) on synthetic datasets. Lower is better.

Method	SEA	HYP	AGR	RBF	Mean
ADWIN	45.2	78.3	52.1	81.4	64.3
DDM	38.7	62.5	48.9	65.2	53.8
EDDM	52.3	71.8	61.4	74.6	65.0
HDDM-A	43.8	75.1	50.3	78.6	61.9
HDDM-W	41.5	58.2	45.7	62.8	52.1
Page-Hinkley	46.9	80.5	53.8	83.2	66.1
KSWIN	49.1	68.7	58.2	71.4	61.9
MDDM	39.4	55.8	44.2	60.5	50.0
ADDM	36.5	51.3	41.6	57.8	46.8
PUDD	32.8	45.2	37.5	51.6	41.8
Meta-ADD	35.2	48.6	39.8	55.3	44.7
AutoDD	**28.4**	**42.1**	**34.2**	**48.7**	**38.4**

TABLE VII: AutoDD proxy-oracle selection accuracy (%) at different stream positions.

Dataset	0–25%	25–50%	50–75%	75–100%
SEA	71.2	85.4	91.8	94.3
HYP	68.5	82.1	89.6	93.1
AGR	65.8	79.3	87.2	91.5
Elec	72.4	84.7	90.3	92.8
Poker	69.1	81.6	88.9	93.6
Mean	69.4	82.6	89.6	93.1

H. Convergence Analysis

Table VII shows how quickly AutoDD learns effective detector selections as the stream progresses, measured as the percentage of windows where the selected detector matches an offline proxy oracle under the same reward used for bandit feedback. AutoDD achieves 69.4% proxy-oracle selection accuracy in the first quarter of the stream, improving to 82.6% in the second quarter, 89.6% at mid-stream, and 93.1% by stream end. This trend suggests that AutoDD learns increasingly reliable selections throughout the stream, while the metric should be interpreted as proxy-oracle agreement rather than direct proof of true drift-label optimality.

VI. Discussion and Limitations

We highlight several boundaries of the current design. First, AutoDD automates detector and hyperparameter selection but not the sliding-window size w, which the sensitivity analysis shows to be the most influential setting; $w = 100$ is robust across our benchmarks, yet selecting it for an unseen stream remains a manual step that an outer adaptive loop could close. Second, the reward is an accuracy-based proxy and therefore entangles detector quality with the downstream learner; we

mitigate this through window-level smoothing and the significance threshold (Section III), but a fully label-free reward is an open problem. Third, to isolate detection from model selection we fix the base learner to GaussianNB and HAT, which are two standard but limited families, and adapt via full model reset, while richer learners and softer adaptation strategies (warm-starting, selective parameter resets, replay, ensemble updating) are left to future work. Finally, AutoDD targets tabular data streams, the prevalent setting in the drift-detection literature; non-tabular modalities such as image, audio, and text are outside the scope of this work.

VII. Conclusion and Future Work

We propose AutoDD, a novel framework that automatically selects and configures drift detectors for data streams. The framework extracts comprehensive meta-features and employs a two-level hierarchical selection strategy using LinUCB for detector selection and Thompson Sampling for hyperparameter configuration. Under the stated proxy-reward and linearity assumptions, the theoretical analysis gives sublinear proxy-regret bounds for the selection layer. Extensive experimental results on synthetic and real-world datasets show that AutoDD achieves consistent mean-accuracy improvements over fixed detector baselines and the compared state-of-the-art methods while requiring no expert intervention. Beyond the limitation-driven extensions discussed in Section VI, a further direction is multi-objective selection that jointly balances predictive accuracy, detection latency, and computational efficiency.

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TABLE VIII: Accuracy (%) by drift type (GaussianNB).

Method	Sudden	Gradual	Incremental	Recurring
ADWIN	82.4	71.3	68.5	75.2
DDM	79.8	74.6	72.1	71.8
EDDM	75.2	78.4	74.8	69.3
HDDM-A	81.7	72.1	69.8	74.5
HDDM-W	80.1	76.2	73.5	74.6
Page-Hinkley	81.9	70.8	68.2	75.8
KSWIN	76.5	77.8	75.2	72.1
MDDM	81.3	75.8	74.1	73.9
ADDM	82.9	77.2	75.6	74.8
PUDD	82.1	79.5	76.8	75.4
Meta-ADD	84.6	**82.3**	78.9	77.8
AutoDD	**88.2**	81.7	**80.4**	**79.6**

TABLE IX: Detector selection frequency (%) by stream characteristic.

Detector	High Velocity	High Noise	High Dim.
ADWIN	28.4	12.3	15.7
DDM	22.1	8.5	11.2
EDDM	8.3	18.7	14.5
HDDM-A	15.2	10.4	12.8
HDDM-W	12.6	14.2	13.1
Page-Hinkley	5.8	8.1	7.2
KSWIN	7.6	27.8	25.5

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APPENDIX A
ADDITIONAL RESULTS

A. Per-Drift-Type Performance

Table VIII shows AutoDD's performance per drift type. AutoDD excels on sudden (88.2%), incremental (80.4%), and recurring (79.6%) drift. Meta-ADD slightly outperforms on gradual drift (82.3% vs. 81.7%) due to its learned drift-type representations.

B. Detector Selection Patterns

Table IX shows that AutoDD's learned selection policy aligns with domain knowledge: ADWIN and DDM are preferred for high-velocity streams, while KSWIN dominates for high-noise and high-dimensional streams.

APPENDIX B
META-FEATURE FORMULATIONS

This appendix gives the full mathematical formulation of each of the 15 online meta-features that form the context vector ϕ_t. Table X lists, per category (statistical, stationarity, and temporal), the closed-form expression or online estimator used to compute every feature over the sliding window of size w.

TABLE X: Meta-feature portfolio with mathematical formulations.

Category	Feature	Formula/Description
Statistical	Velocity	$v_t = n_t/\Delta t$
	Dimensionality	d
	Sparsity	$\frac{1}{wd}\sum_{i=1}^{w}\sum_{j=1}^{d}\mathbb{I}[\mathbf{x}_{i,j}=0]$
	Noise level	$H(\mathbf{X}) \approx -\frac{1}{w}\sum_{i=1}^{w}\log\hat{p}_k(\mathbf{x}_i)$ (k-NN, $k=5$)
	Class imbalance	$\max_c \lvert C_c\rvert / \min_c \lvert C_c\rvert$
Stationarity	Error volatility	$\sigma_{\text{err}}(t) = \sqrt{\frac{1}{w}\sum_{i=t-w}^{t}(e_i-\bar{e})^2}$
	KL divergence	$D_{\text{KL}}(P_t \Vert P_{t-1})$ (histogram, $\sqrt{w}$ bins)
	Autocorrelation	$\rho_t = \frac{\text{Cov}(e_t, e_{t-1})}{\sigma_{e_t}\sigma_{e_{t-1}}}$
	Mean shift	$\Delta\mu_t = \frac{1}{d}\sum_{j=1}^{d}\lvert\mu_{t,j}-\mu_{t-1,j}\rvert$
	Variance shift	$\Delta\sigma_t^2 = \frac{1}{d}\sum_{j=1}^{d}\lvert\sigma_{t,j}^2-\sigma_{t-1,j}^2\rvert$
Temporal	Trend	$\beta_{\text{trend}} = \frac{\text{Cov}(t,e_t)}{\text{Var}(t)}$
	Periodicity	$f_{\text{dom}} = \arg\max_f \lvert\mathcal{F}\{e_t\}(f)\rvert$
	Change frequency	$\frac{1}{\Delta t}\sum_{i=1}^{w}\mathbb{I}[\lvert\Delta e_i\rvert > \tau_{\Delta e}]$
	Change magnitude	$\frac{1}{\lvert\mathcal{C}_t\rvert}\sum_{i\in\mathcal{C}_t}\lvert\Delta e_i\rvert$
	Recurrence score	$\frac{1}{w^2}\sum_{i,j}\mathbb{I}[\lVert\mathbf{x}_i-\mathbf{x}_j\rVert<\epsilon]$